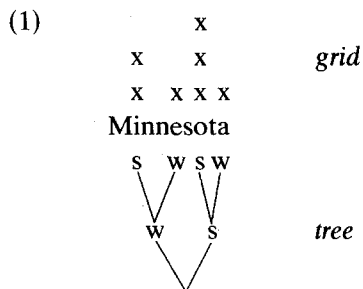


Much of the work on phonology in the last decade has led to an enriched phonological representation incorporating various embellishments on the monolinear string of feature matrices advocated by Chomsky and Halle (1968). One stream of research has produced a collection of proposals that might, in general, be termed “metrical theory”. These proposals share the claim that degrees of stress should be represented in terms of a hierarchical structure of some sort, rather than in terms of phonological features in the string of feature matrices.

For example, Liberman and Prince (1977) propose that stress be represented in terms of two parallel hierarchical representations: the metrical tree and the metrical grid. The nodes of a metrical tree in the theory of Liberman and Prince are maximally binary and labelled pairwise for the relative stress of the material dominated. Strong nodes (labelled "s") dominate material that has greater stress than the material that the corresponding weak node (labelled "w") dominates. The metrical grid is composed of a string of columns where the relative height of columns corresponds to the relative prominence of the segmental material aligned with the grid. An example is given in (1) showing how these devices represent the stress pattern of a word like *Minnesota*.



* A much earlier version of this work was presented at *NELS* in 1984. This paper also incorporates material presented at the Milwaukee Symposium on Language Typology and Universals 1985. Various drafts owe much to the following individuals: D. Archangeli, S. Davis, J. Goldsmith, B. Hayes, J. Levin, E. Moravcsik, A. Prince, D. Pulleyblank, L. Rousos, and five anonymous reviewers. All errors of data or analysis are the author's responsibility.

In the grid, the vowel with primary stress, *o*, has the tallest grid column with three xes. The vowel *i* with tertiary stress has the next highest grid column with two xes.¹ Lastly, *e* and *a* with no stress have grid columns of one each. In the metrical tree, the vowels *e* and *a* are immediately dominated by weak nodes and are the weakest in the tree. The vowels *i* and *o* are both labelled strong and have stronger stress than *e* or *a*. The strong node higher in the hierarchy (lower on the page) indicates that *o* is stronger than *i*.²

One of the central debates in metrical theory is whether two parallel representations are necessary. Some have argued that the tree is unnecessary and that the metrical grid is a sufficient and appropriate representation for phonological stress, e.g., Prince (1983) and Selkirk (1984). Others have argued that the grid is unnecessary and that the metrical tree is the proper representation, e.g., Hammond (1984) and Hayes (1985).³ This debate is not relevant to the present paper; I simply assume that the grid is unnecessary, and that the proper representation for stress is a tree-like structure. Additional arguments in favor of a similar representation can be found in Halle and Vergnaud (1985).

The metrical theory of Liberman and Prince has undergone other changes too. One important development is the introduction (from classical poetic scansion) of the METRICAL FOOT, which dominates one or more syllables. Hayes (1981) suggested that there are three kinds of feet with respect to their sensitivity to syllable weight. First, there are quantity-sensitive feet. These are feet where the nonheads, the weak terminals of (1), may not dominate heavy syllables, that is, syllables containing a long vowel and/or a final consonant.⁴ Second, there are quantity-insensitive feet. These are feet where syllable weight is irrelevant. Last, there are obligatory-branching (OB) feet, the type focused on in this paper. These are feet where the heads, the terminal *s* nodes of (1), must dominate heavy syllables, and the nonheads may not dominate heavy syllables. The term "obligatory branching" derives from the assumption, irrelevant here, that heavy syllables have a branching structure. These three foot types can be represented in chart-form as in (2), using two

¹ Traditionally, this is called a secondary stress. In the SPE system, it is a tertiary. The crucial fact is that it is weaker in stress than *o*.

² Reading stress off trees and grids is actually more complex than this. This brief discussion is not intended to present the intricacies, but only to give the rough outlines of the procedure. For a complete account, see Liberman and Prince (1977).

³ For additional arguments going both ways, see, for example, Prince (1985), Hayes (1984), Rappaport (1984), Bengier (1984), Al-Mozainy, Bley-Vroman, and McCarthy (1985).

⁴ See below for exemplification. Here we only sketch out the main thrust of our argument.

binary choices:

(2)		[-H]	[+H]	QI: quantity-insensitive
				QS: quantity-sensitive
				OB: obligatory-branching
	[-N]	QI	?	[+H]: restricted head
	[+N]	QS	OB	[+N]: restricted nonhead

One choice, $[\pm H]$, governs whether the heads of feet can dominate only heavy syllables (+) or may dominate any kind of syllable (-). This allows us to distinguish quantity-insensitive and quantity-sensitive feet from obligatory-branching feet. The other choice, $[\pm N]$, governs whether nonheads are prevented from dominating heavy syllables (+) or are not so restricted (-). This allows us to distinguish quantity-insensitive feet from quantity-sensitive feet and obligatory-branching feet.

Note that this categorization of foot types results in an empty cell in (2). There are no feet where the head must dominate a heavy syllable and nonheads can dominate any kind of syllable. The question addressed in this paper is whether this is where the gap lies. I will argue that there is a more natural foot typology that is empirically superior to the one in (2). Specifically, I argue that there are no $[+N, +H]$, (OB) feet. Rather, there are feet which correspond precisely to “?” in (2): the head must dominate a heavy syllable but nonheads are unrestricted. Data will be presented that argue for $[+H, -N]$ feet, what I term “revised obligatory-branching” (ROB) feet. First, I show that ROB feet can account for all the facts accounted for with OB feet, using data from Turkish and Khalkha Mongolian. Next, I show that ROB feet can account for additional data drawn from the stress systems of Lenakel and Klamath, which OB feet cannot handle.

Before showing the empirical advantages of ROB feet over OB feet, I will consider the conceptual consequences of the typology to be proposed. The system is given in (3) below.

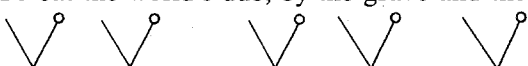
(3)		[-H]	[+H]	QI: quantity-insensitive
				QS: quantity-sensitive
				OB: obligatory-branching
	[-N]	QI	ROB	[+H]: restricted head
	[+N]	QS		[+N]: restricted nonhead

The conceptual superiority of (3) over (2) lies in the interpretation of the ‘features’ $[\pm N]$ and $[\pm H]$. These ‘features’ share the property that a

plus indicates some degree of restrictiveness. With [+H] feet, the heads are restricted, and with [+N] feet, the nonheads are restricted. Viewed in this light, the asymmetry noted in the system in (2) is curious. Why is there no foot type where just the heads are restricted: [+H, -N]? The gap in (3), however, can be regarded as a foot type with too many restrictions. The empty cell represents a foot type in which both heads and nonheads are restricted – as was the case with the original OB foot in which heads must dominate heavy syllables, nonheads light syllables. I claim that such feet do not occur, being too heavily restricted.

1. BACKGROUND

Metrical theory uses some concepts of classical poetic meter to describe stress systems. The particular notion drawn from poetic meter is the metrical foot, a prosodic constituent taken as the building block of poetic verse. In (4), I exemplify a line of iambic pentameter. Such a line is built of five binary feet, where the right node of each foot is designated strong, indicated here with a small circle and a vertical line.

- (4) To éat the wórl'd's due, bý the gráve and thée
- 

In (5) is a line of trochaic tetrameter.

- (5) Gítche Mánitó, the míghty
- 

This line type is constructed of four binary feet, where the left node of each foot is strong.

Metrical theory exploits these foot types – iambic (right-headed) and trochaic (left-headed) – in representing stress patterns. English (Lieberman and Prince, 1977) has left-headed feet as in (6) and Modern Hebrew (Bolzky, 1982) has right-headed feet as in (7).

- (6) àpalàchicóla
- 

- (7) mevùgarím 'adults'



The circled nodes of feet – the heads – indicate stressed syllables and the nonheads (uncircled) indicate unstressed syllables.

This, of course, is only part of the story. We must be able to distinguish primary stresses from secondary stresses, and we must account for how stresses are assigned to strings, that is, how the trees get there.

Hayes (1981) proposes a theory of stress assignment that defines an interesting typology of stress systems. Oversimplifying temporarily, stress assignment has four parameters, italicized in (8), and is divided into three steps: (8a), (8b), and (8c).

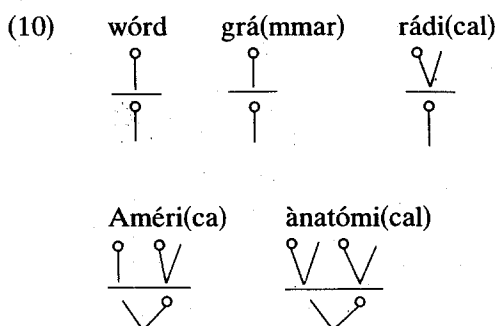
- (8)a. *Extrametricality*: A peripheral syllable, either the leftmost or rightmost, may be excluded from the scansion.
- b. *Foot-Headedness*: Every foot has a head, which defines the stressed element. Right-headed feet have the head on the right, and left-headed feet have the head on the left.
Foot-Directionality: Feet can be built from left to right or from right to left.
- c. *Word Tree Headedness*: An unbounded left-headed or right-headed structure may be built on the roots of feet.

First, there are rules of extrametricality. A rule of extrametricality can remove a peripheral syllable in the string from the domain of stress assignment. Second, there is foot construction. Basically, foot construction is determined by setting two parameters: foot-headedness and foot-directionality. Foot-headedness tells us whether the feet constructed have the head on the left or on the right. Foot-directionality tells us whether foot construction starts at the left edge of each word in the string or the right edge. Lastly, word tree construction determines which foot is to bear the main stress. The word tree is an *n*-ary branching structure built on the roots of feet. Either the rightmost or leftmost node of the word tree is the head. The head of the word tree dominates the main stress.

To see how this works, consider the preliminary analysis of English given in (9). This is exemplified in the examples in (10). The basic pattern

is antepenult main stress with secondaries iterating leftward from the primary.⁵

- (9)a. The final syllable is extrametrical in a polysyllabic word.
- b. Left-headed feet are built right to left.
- c. Build a right-headed word tree.



First, the rightmost syllable of polysyllabic words is made extrametrical. Extrametrical syllables are indicated with parentheses. Next, left-headed feet are built from right to left. That the feet be left-headed is required if words like *radical* and *anatomical* are to be properly stressed. That the feet be built from right to left is required if *America* is not to be stressed **Àmerica*. Notice that, if in the course of foot construction only a single syllable is available for footing, then the foot has only one node, the head, as in *word*, *grammar*, and the second foot constructed in *America*. Such feet are termed “degenerate”. Lastly, a right-headed word tree is constructed. This has the consequence of making the rightmost stress the primary.

A second language illustrating these parameters is Pintupi, a Pama-Nyungan language of Australia (Hansen and Hansen, 1969). The examples in (11) exemplify the basic pattern. The first syllable bears the primary, and alternating following syllables receive secondary stresses, except that the ultima is never stressed.

- (11) páṇa ‘earth’
 tyútaya ‘many’
 málawàna ‘through (from) behind’
 púṭṭkàlatju ‘we (sat) on the hill’
 tílirṇulàmpatju ‘the fire for our benefit flared up’

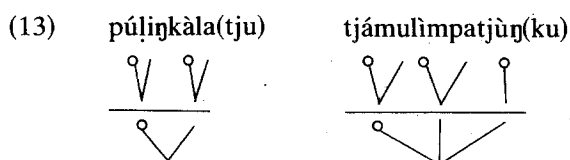
⁵ Stress in English is much more complicated than this. See Chomsky and Halle (1968) for the details. See Liberman and Prince (1977), Hayes (1981), and Hammond (1984) for metrical treatments.

Since the alternation seems to be determined from the left edge of the word, foot construction must be left to right. Since the first syllable has stress, the feet must be left-headed. Since the first stress is the main stress, the word tree is left-headed. Lastly, one way to keep stress off the ultima is by making it extrametrical.

The analysis in (12) yields this pattern.

- (12)a. Make the final syllable extrametrical.
- b. Build left-headed feet left to right.
- c. Build a left-headed word tree.

This gives sample trees as in (13).



The stress assignment algorithm defines a finite number of tree types. Since I have presented four binary parameters, in (8), that allows for sixteen different possibilities. In fact, there are additional parameters that are independently variable and that I now consider. These are: QUANTITY-SENSITIVITY, BOUNDEDNESS, and the OBLIGATORY-BRANCHING parameter.

I begin with what Hayes called "quantity sensitivity". In Pintupi and the subset of English presented thus far, the feet constructed appear to be insensitive to syllable weight. That is, the footing process just assigns the appropriate structure regardless of the segmental content of the syllables. In some languages, segmental makeup of syllables is relevant to the scansion. In particular, in certain languages, heavy syllables or long vowels attract stress. By heavy syllable is meant a syllable closed by a consonantal coda or containing a long vowel. In fact, when English is considered more closely, it is apparent that heavy syllables attract stress. The facts are evident in the following array of data (from Chomsky and Halle, 1968). The first column contains words with light penults. The second and third columns have words with heavy penults. In column two, the penults are heavy because they are closed. In column three the penults are heavy because they contain long vowels. Only the first column can be accounted for by the analysis in (9) above. In the second and third columns, that analysis predicts incorrect forms like **vérandā* and **ároma*.

(14)	<i>light</i>	<i>heavy</i>	
		<i>closed</i>	<i>long</i>
	cínema	veránda	aróma
	jávelin	synópsis	hiátus
	vénison	amálgam	horízon
	ásterisk	asbéstos	aréna
	ársenal	agénda	factótum
	lábyrinth	phlogíston	coróna

To account for the forms in columns two and three, we must somehow require that the first foot constructed be degenerate just in case the penult is heavy. That is, we want the output of footing to be (15) rather than (16), so that a right-headed word tree will place stress on the heavy penult.

(15)	veran(da)	aro(ma)
	◊ ◊	◊ ◊

(16)	veran(da)	aro(ma)
	◊	◊

In our terms, some languages stipulate that nonheads may not dominate heavy syllables. This modifies the foot construction algorithm. Maximally binary feet in English are built subject to the quantity-sensitivity requirement. If by building a binary foot, quantity-sensitivity would be violated, a degenerate foot is built instead. This guarantees that for English the output of footing is (15) rather than (16) and that the correct results are achieved for columns two and three of (14).

Now consider the boundedness parameter. Thus far, all the languages considered here have feet that are maximally binary, or bounded. Certain languages exhibit feet that are maximally unbounded. A language exhibiting unbounded feet is Turkish (Kaisse, 1985; Sezer, 1983; Underhill, 1976).⁶ Stress, in the normal case, always falls on the ultima.

(17)	odá	'room'	taní	'know'
	odalár	'rooms'	tanıdık	'acquaintance'

⁶ We ignore for now unstressable suffixes and lexically accented stems. An anonymous reviewer points out that these facts from Turkish are consistent with an analysis in terms of bounded feet. In fact, Prince (1985) argues that all instances of unbounded feet can be eliminated in favor of bounded binary feet. This fact is largely irrelevant to the arguments developed in this paper, and we therefore assume an unbounded analysis.

To account for this, the parameter of *boundedness* has been proposed. Bounded feet, as in Pintupi and English, are maximally binary. Unbounded feet, as in Turkish, are maximally *n*-ary. One foot will cover the entire span, e.g.,

- (18) odalár


(Notice that, in such a case, the foot may be built left to right or right to left. Unboundedness obviates directionality if footing is exhaustive.)

There are languages where unboundedness and quantity-sensitivity operate together. An example illustrating this is Waalubal (Crowley, 1978) a dialect of the Bandjalang language of Australia. The following data exemplify the basic pattern. The generalization is that the first syllable always bears main stress. All long vowels in the word bear secondary stress.

- (19) bándan 'other'
 bá:nbini 'took out'
 ɲámà:lu 'tree goanna erg.'
 wúrgulù:m 'magpie'
 bándanibè: 'only covered'

Notice that there is no binary counting involved. If stress were assigned by means of bounded binary feet, then one would expect different stress patterns in odd-syllabled words (e.g., **wúrgulù:m**) and even-syllabled words (e.g., **bándanibè:**). Since that is not the case, stress is assigned by a different theoretical device; in this case, quantity-sensitive unbounded feet.

Quantity-sensitivity, as we saw in the English example, is the stipulation that various kinds of heavy syllables cannot be nonheads. In Waalubal, since long vowels seem to attract stress, it must be the case that long vowels cannot be nonheads. The feet must be unbounded, as large as the domain they are assigned to, provided the nonheads do not violate the quantity-sensitivity stipulation. This means that the only syllables that get stress are long vowels, and one of the word edges. Which word edge is determined by the headedness of the feet.

To assign stress to the first example **bándan**, the feet constructed must be left-headed, as in (20).

- (20) bándan


This also works for **bá:ɲbini**. Notice though that the feet must necessarily be unbounded to account for **bá:ɲbini** as in (21). If they were bounded, that is, binary, one of the ungrammatical forms in (22) would result.

- (21) **bá:ɲbini**
- 

- (22) ***bá:ɲbini** ***bá:ɲbini**
- 

Notice also that if there are no noninitial long vowels, then only one foot will be built. This is because the feet are unbounded. They are as large as the domain, provided the nonheads do not dominate long vowels. That result is guaranteed as long as there are no noninitial long vowels.

Consider now what happens when there is a noninitial long vowel as in **ɲámà:lu**. Given exhaustive footing with left-headed feet, there are four logical possibilities.

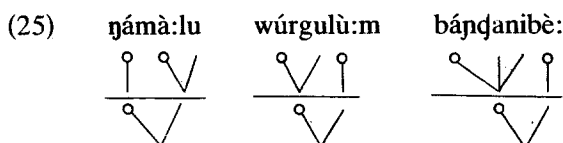
- (23)a. **ɲámà:lu** b. **ɲámà:lù**
- 
- c. **ɲámà:lù** d. **ɲámà:lu**
- 

Of these four, (a) and (b) are excluded because the quantity-sensitivity stipulation is violated, while (c) is excluded because of the maximality aspect of foot construction: feet must be as large as possible subject to the length of the domain and quantity-sensitivity. Since the last two syllables of **ɲámà:lu** *can* be footed together, they *must* be footed together.

The last two examples work like **ɲámà:lu**. Their tree structures are given below in (24).

- (24) **wúrgulù:m** **báɲɲanibè:**
- 

To guarantee that the first stress of the word is the main stress, the word tree must be left-headed. This gives results as below for words with more than one foot.



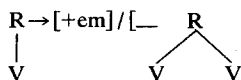
The complete analysis of Waalubal is as follows. Feet are left-headed, unbounded, and quantity-sensitive to length. The word tree is left-headed.⁷

In partial summary, the version of metrical theory that we are adopting here represents phonological stress in terms of tree structures assigned to the string. The algorithm for assigning tree structures allows for parametric variation at six points: extrametricality, foot-headedness, foot-directionality, word-tree-headedness, quantity-sensitivity, and boundedness. In addition, it has been proposed that the algorithm include a parameter called the obligatory-branching parameter (OB). In the following section, we consider what this is meant to account for.

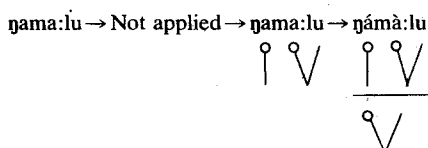
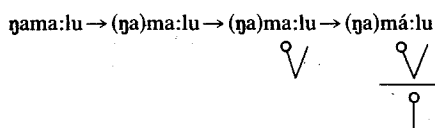
2. OBLIGATORY-BRANCHING

In this section, we review Hayes' (1981) analyses of Eastern Cheremis and Khalkha Mongolian and Kaisse's (1985) analysis of exceptional

⁷ Crowley claims that in words with an initial short vowel followed immediately by a long vowel, the initial syllable may be unstressed with main stress occurring on the second syllable: $\eta\acute{a}m\acute{a}:lu$ is an optional variant of $\eta\acute{a}m\grave{a}:lu$. To account for this, I propose that stress assignment be preceded by an optional extrametricality rule. That rule, given below, optionally makes an initial short vowel extrametrical when followed immediately by a long vowel.



This will give the following two derivations for $\eta\acute{a}m\acute{a}:lu$:



To get stress on the first long vowel in the span, one could build a right-headed quantity-sensitive foot on the left margin. This gets the right results in examples like the following.

- (29) garáasaa bariáad
  

However, this leads us to expect final stress in the event there is no long vowel in the span. This is incorrect.

- (30) *alí *xõtəlbərə́
  

The correct stress patterns for the items in (30) could be generated if one built a left-headed quantity-sensitive foot on the right margin.

- (31) áli xõtəlbərə
  

This, however, predicts stress on the last of a sequence of long vowels, which is also incorrect.

- (32) *garaasáa
 

Thus Hayes' notion of quantity-sensitivity entails something of a paradox for languages like Khalkha. To remedy this, Hayes adopts Halle and Vergnaud's (1978) OBLIGATORY BRANCHING parameter (OB). Under OB footing, heads *must* dominate heavy syllables (long vowels) and nonheads *must not* dominate heavy syllables (long vowels).

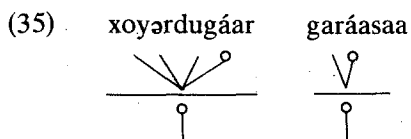
In Khalkha, a right-headed OB foot sensitive to vowel length is built on the left margin. This produces examples like the following.

- (33) xoyərdugáar garáasaa
  

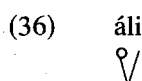
If the word contains no long vowel, then no foot is built, since no footing results in the head dominating a long vowel.

- (34) *ali *ali
  

The output of foot construction is incorporated into a left-headed word tree. In cases like (33), this has no consequences.



But in cases like (34) the effect of word tree construction is essential. If there is no foot, word tree construction takes syllables as terminals.



This stress assignment procedure thus accounts for stress systems like Khalkha Mongolian, as opposed to Eastern Cheremis.

The OB analysis of Khalkha exploits unbounded feet. There are also OB analyses that exploit bounded or binary feet. Kaisse (1985), for example, describes the stress pattern of certain exceptional words in Turkish in terms of binary OB feet. As described above, (see (17)), stress in Turkish normally falls on the ultima. However, some place names and other borrowed words are stressed in a different fashion. Kaisse cites the data in (37)–(39) from Sezer (1983). The data in (37) show that words with heavy penults get penult stress.

(37)	Samuélson	(Paul) ‘Samuelson’
	Vasínkton	‘Washington’
	Antálya	city in Turkey
	lokánta	‘restaurant’
	Istánbul	city in Turkey
	Ayzinhó:ver	‘Eisenhower’

The data in (38) show that if the penult is light and the antepenult heavy, then the antepenult is stressed.

(38)	Ánkara	city in Turkey
	pénjere	‘window’
	samándıra	‘buoy’
	šévrole	‘Chevrolet’

If both penult and antepenult are light, then the penult is stressed.

- | | | |
|------|----------|----------------|
| (39) | Kenédi | 'Kennedy' |
| | Pitolémi | 'Ptolemy' |
| | jubíle | 'jubilee' |
| | Göréme | city in Turkey |
| | Adána | city in Turkey |

To account for the fact that stress always falls on the penult or antepenult, Kaisse makes the final syllable extrametrical, and builds a binary foot at the right edge of the word. Since syllable weight is clearly relevant to the scansion, the foot must be either QS or OB. The foot cannot be QS, because either headedness produces the wrong results for one of the three cases above. In (40), I show the result of a right-headed binary QS foot applied to all combinations of syllable weight involving penult and antepenult, and in (41) I show left-headed binary QS footing. Notice the incorrect (40c) and (41d).

- (40)a. Antál(ya) b. lokán(ta) c. *Anká(ra) d. Kené(di)
- 




- (41)a. Antál(ya) b. lokán(ta) c. Ánka(ra) d. *Kéne(di)
- 




Therefore the feet constructed must be OB. Kaisse proposes a left-headed binary OB foot.⁹

- (42)a. Antál(ya) b. lokán(ta) c. Ánka(ra) d. Kene(di)
- 



Since the foot is OB, no foot is constructed for **Kenédi**, because neither penult nor antepenult is heavy. To remedy this, one might propose that exceptional words get the unbounded right-headed foot that normal unexceptional words get. This results in penult stress for **Kenédi**, since the final syllable is extrametrical.

⁹ In an earlier form of this paper, I followed Prince (1983) in assuming that all apparent cases of binary OB could actually be reanalyzed in terms of extrametricality. This is because none of the cases in the literature involved both OB and extrametricality, as Turkish appears to. The Turkish example cannot be easily reanalyzed using extrametricality. Therefore, I follow Kaisse here in her argument that Turkish constitutes a counterargument to the grid-based theory of stress outlined in Prince (1983).

- (43) Kenedi → Kene(di) → Exceptional OB Footing N/A →
 Regular → Kené(di)
 Footing 

This analysis shows that in addition to unbounded OB feet, there need to be bounded OB feet.

3. REVISED OBLIGATORY BRANCHING

In this section, I show how Khalkha Mongolian and Turkish can be analyzed using ROB feet. First consider stress in Khalkha. Recall that the generalization in Khalkha was that stress falls on the leftmost long vowel in the word (see (28)). If the word contains no long vowels, then the first syllable gets stress. Using OB feet, this called for an unbounded right-headed OB foot. The relevant algorithm seeks out the leftmost heavy syllable. In (2) and (3) above, I suggested that it would be conceptually superior to replace OB feet with feet where only the heads are restricted, and the nonheads can dominate heavy syllables (in this case, long vowels): ROB feet.

If the right-headed OB foot in Khalkha were replaced with a right-headed ROB foot, then the foot-forming algorithm would seek out the rightmost long vowel. That is because an ROB foot may dominate any number of long vowels: the nonheads are unrestricted. A form like **garáasaa** would get footed as in (44).

- (44) *garaasáa


Since a right-headed ROB foot picks out the rightmost long vowel, a left-headed foot will pick out the leftmost long vowel. With this in mind, let us revise our analysis above accordingly.

- (45)a. Build an unbounded left-headed ROB foot.
 b. Built a left-headed word tree.

This entails derivations like the following for words like **xó̌təlbəɾə**.

- (46) xó̌təlbəɾə → N/A → xó̌təlbəɾə


The left-headed word tree takes syllables as terminals since no ROB foot can be built.

Considering forms like **garáasaa** again, derivations like the following must be prevented.

(47) garaasaa → garáasaa → *gáráasaa

This can be prevented by adjoining unfooted syllables before word tree construction. This is exemplified in (48) below.

(48) garaasaa → garáasaa → garáasaa

Notice that applying stray adjunction prior to word tree construction has no consequences elsewhere in the stress system. It is only relevant in the case of OB or ROB feet.

We now show how ROB feet can replace OB feet in the analysis of exceptional stress in Turkish. As in the analysis of Khalkha, we need only reverse the headedness of the OB foot in order to come up with an analysis in terms of ROB feet. Recall that the generalization in Turkish was that stress falls on the antepenult if it is heavy and the penult light; otherwise, stress falls on the penult. Above, we gave the following analysis.

- (49)a. Make the final syllable extrametrical.
 b. Build a binary left-headed OB foot on the right edge.
 c. If no foot is built, build an unbounded right-headed foot.

Revising this in terms of ROB feet, we merely replace “left-headed OB foot” with “right-headed ROB foot.” Applying (49a) and the revised form of (49b) to the examples in (40), (41), and (42), we get the following.

(50)a. Antál(ya) b. lokán(ta) c. Ánka(ra) d. Kene(di)

Notice that no ROB foot is built in (50d), because neither of the metrical syllables is heavy. Subsequent unbounded footing (49c) provides the penult stress.

(51) Kenedi → Kene(di) → ROB N/A → (49c) → Kené(di)



Notice also the results of ROB footing in the case of (50c). One may have expected no ROB foot here as well. However, a degenerate foot is built, since the antepenult is heavy. Rather than seem odd, this result is actually quite in keeping with the way QS foot construction is applied to a span. See the discussion of examples (15) and (16) above. There we suggested that a degenerate foot is built if the QS stipulation would be violated by building a binary foot. Here, in (50c), we say the same thing: a degenerate foot is built if the ROB stipulation is not met. If it still is not met, build no foot, as in (50d).

We have shown that languages like Khalkha and Turkish can be accounted for using ROB feet. This required that stray syllable adjunction apply before word tree construction, a move with no empirical consequences beyond ROB feet. In addition, we showed that the procedure for building ROB feet is closely similar to the procedure for building QS feet. We now turn to data from two languages, Lenakel and Klamath, which suggest that ROB feet are empirically superior to OB feet.

4. LENAKEL

In (52) the stress pattern of Lenakel nouns is exemplified.¹⁰ Stress appears to iterate from the right edge of the word. Even-numbered syllables from the right are stressed. The rightmost is primary.

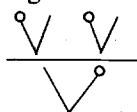
(52)	n ám <i>N fish</i>	'fish'	kÓIEy	'sweet potato'
	kayÉyaw	'hat'	l àgabnábən <i>in morning</i>	'(in the) morning'
	kayÈlawÉlaw	'kind of dance'	lÈ duḃèlugálUk ^h <i>in lungs</i>	'lungs' (LOC)

To get this stress pattern, left-headed feet are built leftward and in-

¹⁰ In Lynch (1974) the facts to be presented here are given an SPE-type analysis. In these data, [v] is a high central glide, capital letters indicate lax vowels, and labialized labials have been transcribed with macrons. Certain allophonic processes are suppressed. For details on the allophony, see Lynch (1974, 1978). Some of the forms to be presented exhibit considerable internal structure. Abbreviations in the morpheme glosses: CUR: current, SG: singular, PL: plural, EX: exclusive, DU: dual, CONJ: conjunctive, PERF: perfective, CONT: continuative, FUT: future, INTERR: interrogative, 1: first person, 2: second person, 3: third person, PX: possessor prefix, TR: trial.

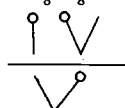
corporated in a right-headed word tree. For example:

(53) làgabnìbən



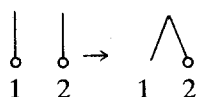
This accounts directly for the examples with an even number of syllables, but gives the wrong results for examples with an odd number of syllables.

(54) *kàṽÉṽaw



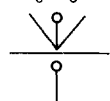
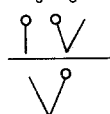
To account for cases like this, I assume a rule with the structural change given in (55).¹¹

(55) *Destressing*



This rule applies to remove a foot in a stress clash and adjoin it to the following foot. It produces derivations like (56).

(56) kàṽÉṽaw → kàṽÉṽaw → Destressing → kàṽÉṽaw



Compare the verbal stress pattern. Here, stress appears to be iterating from two directions. The primary always falls on the penult if there is one. The secondaries fall on every other syllable counting from the left edge of the word.

(57) r Ós
3SG take
'he took it'

¹¹ As noted above, we sidestep the theory of rhythm and destressing rules presented in Hammond (1984). However, the interested reader can note that all such rules proposed here are in conformity with that proposal. The details of it would take us far afield, so rules like (55) are adopted as descriptive expedients.

ár Ow

PL blind

'to follow'

y ag yá gən

1PLEX CUR DU eat

'we eat it'

n ìm alìwOk^h

2SG CONJ walk

'you walked'

k ən am ar gənìm

3PL PERF CONT PL pinch

'they have been pinching it'

t y àg am àr OlgÉygEy

FUT 1PLEX CUR CONT PL like

'we will be liking it'

t ìn ag àm ya sɪŋóvɪn

FUT 2 CUR CONT DU copy

'you will be copying it'

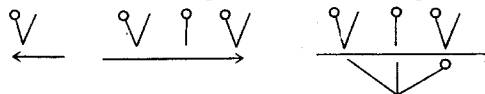
nà d y ag àm Edw àdamníṁēn

about-to FUT 1PLEX CUR CONT INTERR shake

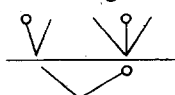
'why am I about to be shaking'

To account for these, I propose that foot construction is bidirectional for verbs. First, one foot is built on the right margin. Then feet are built from the left margin. As with nouns, all feet are left-headed. The output of foot construction is incorporated into a right-headed word tree and undergoes Destressing (55). Following is a sample derivation.

(58) kənamargənìm → kənamargənìm → kənamargənìm



→ Destressing → kənamargənìm



Notice how bidirectional foot construction creates the context for destressing on the antepenult in all cases with an odd number of syllables.¹²

Lenakel exhibits tense and lax vowels in near complementary distribution. High vowels are tense in open syllables and lax in closed ones. Mid vowels are lax before consonants and tense otherwise. The examples in (59) show that nonlow (noncentral) vowels are tense before other vowels and word-finally. The examples in (60) show that nonlow vowels are lax in closed syllables, those in (61) show that high vowels are tense preconsonantly in open syllables, and those in (62) show that mid vowels are lax when followed by a consonant – even in an open syllable. (The vowel [i] is an allophone of /ə/ occurring after a coronal.)

- | | | | | | |
|------|-------------------|----------------|----------------------------|------------------|-----------------------|
| (59) | pía | k ^h | 'my brother' | kádʔi | 'coconut pudding' |
| | <i>brother</i> | <i>ISG</i> | | | |
| | núa | k ^h | 'my shoulder' | álu | 'to dig' |
| | <i>shoulder</i> | <i>ISG</i> | | | |
| | mèaméa | | 'ant' | ní de | 'taro' |
| | | | | <i>SG taro</i> | |
| | kàmadóa | | 'kind of taro' | amágo | 'to dance (of women)' |
| (60) | Ilmáŋa | | 'Eromanga Isl.' | kíŋ | 'kind of worm' |
| | kÚmgUm | | 'to be stiff' | sÚk ^h | 'spear' |
| | Élmas | | 'to frighten' | kát hE1 | 'we' |
| | | | | <i>IIN TR</i> | |
| | vilÓvlo | | 'kind of bird' | tigÓmgOm | 'Branches' |
| (61) | ígo | | 'to be twisted' | avíliŋ | 'to be thin' |
| | úlík ^h | | 'to be hard and dry' | n údo | 'kind of arrow' |
| | | | | <i>SG arrow</i> | |
| (62) | Éna | | 'to suck in one's stomach' | kəmÉlu | 'kind of basket' |
| | Ódʔi | | 'to separate' | kÓlaw | 'kind of tree' |

¹² One anonymous reviewer claims that the facts in (57) are consistent with a lexical theory of bidirectional stress assignment explored in Levin (1982). He proposes that the verb stem and suffixes are stressed at an early stratum by right-to-left footing. After prefixes are added at a later stratum, left-to-right footing applies. A form like *kənamargónim* would then undergo the following derivation. First, right-to-left footing applies to the stem and its suffixes. In this example, there are no suffixes: *gónim* becomes *gónim*. Prefixes are added, and left-to-right footing applies: *kənamargónim* becomes *kənamàrgónim*. Destressing applies as in the text. This approach unfortunately does not work in the shorter bisyllabic and trisyllabic forms of (57). Consider the predicted derivation for *yagyágán*. First, right-to-left footing applies to the stem *gán*. Prefixes are added and left-to-right footing produces **yagyágán*.

To account for this distribution, I adopt the phonemic inventory in (63) and the rule in (64), where /ə/ and /a/ are underlyingly lax and hence unaffected by the rule.¹³

- (63)
- | | |
|---|---|
| i | u |
| e | ə |
| | o |
| | a |

- (64) *Laxing*
-

This rule laxes high vowels in closed syllables, and mid vowels /e/ and /o/ preconsonantly even in open syllables. In (65)–(67), I give some representative examples of the underlying forms that this proposal entails.

- (65) *word-final*

i:	/kad ^y i/	→	[kád ^y i]
u:	/alu/	→	[álu]
e:	/nəde/	→	[níde]
o:	/amago/	→	[amágo]
a:	/meamea/	→	[mèaméa]
ə:	does not occur		

- (66) *preconsonantal in an open syllable*

i:	igo	→	[ígo]
u:	/ulək ^h /	→	[úlík ^h]
e:	/ena/	→	[Éna]
o:	/od ^y i/	→	[Ód ^y i]
a:	/meamea/	→	[mèaméa]
ə:	/kəmelu/	→	[kəmÉlu]

- (67) *in a closed syllable*

i:	/ilmaŋa/	→	[Ilmáŋa]
u:	/kumgum/	→	[kÚmgUm]
e:	/elmas/	→	[Élmas]
o:	/təgomgom/	→	[tigÓmgOm]
a:	/kolaw/	→	[kÓlaw]
ə:	/ulək ^h /	→	[úlík ^h]/ə/ → [ɪ] after a coronal

¹³ We follow Lynch in assuming /ə/ and /a/ are lax. Lynch actually assumes that all the vowels are lax underlyingly. He offers two motivations for this move. First, all vowels are then lax underlyingly. Second, the mid vowel laxing clause of (64) may seem unnatural.

(68) asís 'to swell up' amnúm 'to drink'
abgén 'to be jealous' yElmów 'salt water eel'

(69) *ásIs *ámnmUm
 *ábgnEn *yÉlmOw¹⁴

(70) kəbás 'axe' almál 'to be insane'

(71) *kábas *álmæl

(72) ni gí níl ar 'their hearts'
PX heart 3 PL

One possibility for accounting for these vowels would be to assume that they are lexically stressed. That is, we could assume that underlyingly these vowels have lexical foot structure. The lexical contrast would be as in (73) below.

(73) /abgen/ /kat^hel/ ([kat^hEl])

¹⁴ Note that **w** does close the syllable: **Ow** 'to be blind'.

laxing could be ordered before stress assignment, and revised to incorporate the lexical distinction in (73). The revision necessitated would be that laxing would apply only to (lexically) unstressed vowels. The problem with this approach is that, as will be demonstrated below, there are other cases that must be treated as lexically accented and that (1) do undergo laxing, and (2) behave differently with regard to stress assignment. Since these other cases that behave differently are best represented as lexically stressed, that option is not available to describe these cases. We will also see that the behavior of stress-attracting laxing vowels in sequence argues against the lexical foot approach.

Another strategy would be to set up a phonological distinction that is ultimately neutralized. One possibility along these lines would be to set up an underlying contrast between long vowels and short vowels. Lynch, in fact, makes this proposal, and we will ultimately adopt it below. This would mean that stress-attracting nonlaxing vowels would be represented as underlyingly long, while normal vowels would be represented as underlyingly short. We would have the contrast (74) instead of (73) above.

(74) /abgeen/ /kat^hel/

This would allow us to invoke quantity-sensitivity to get the right stress: long vowels may not be nonheads. In addition, the laxing rule could be revised so that long vowels would not undergo it. This move has the following desirable consequences. First, it allows us to invoke an existing parameter to get the stress: quantity-sensitivity. Second, the restriction proposed for the laxing rule, that it may not apply to long vowels, seems natural inasmuch as other languages have similar restrictions, e.g., English.

This move does have the undesirable property that there is no surface manifestation of length for the stress-attracting nonlaxing vowels: that is, this is a kind of absolute neutralization. Therefore, it is reasonable to explore the alternative of marking the contrast diacritically. For example, as in (75).

(75) /abgen/ /kat^hel/
 [+D] [-D]

One could then make stress, and laxing sensitive to this distinction: [+D] vowels may not be nonhead terminals, and laxing may not apply to [+D] vowels. This approach avoids absolute neutralization, but leaves mysterious why both stress and laxing are sensitive to this feature. In conclusion, we follow Lynch in representing these vowels as underlyingly

long, and add quantity-sensitivity and the laxing revision to our rules as given so far.¹⁵

- (76) *Laxing* (revised)
-

Now Laxing will affect only [-long] vowels.

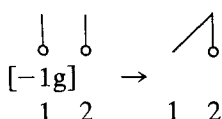
This proposal predicts that long vowels would interrupt the alternating patterns observed above. Though, as noted above, long vowels only occur in morpheme-final position, this is not hard to demonstrate.

- (77) /nɪ gii nil ar/ nɪgɪnɪlar 'their hearts'
PX heart 3PL PL

/nɪ man sii nil ar/ nɪmansɪnɪlar 'their buttocks'
PX ? buttocks 3PL PL

To account for the forms above, one need only assume that Destressing, too, does not apply to long vowels. This revision is given below.

- (78) *Destressing* (revised)



This results in derivations like the following.

- (79) nɪ-gii-nil-ar → nɪgiinilar → nɪgiinilar → Destressing → nɪgɪnɪlar
-

- (80) nɪ-man-sii-nil-ar → nɪmansiinilar → nɪmansɪnɪlar
-

¹⁵ See Anderson (1984) where the kind of move we adopt in Lenakel is suggested for all languages with fixed stress.

In (79), first feet and the word tree are built. Then (78) removes the first foot. In (80), after word tree construction, (78) is inapplicable.

The facts get more complicated when one considers adjacent long vowels. The following forms all contain final long vowels. This accounts for their final stresses.

- (81) /r is gən aan/ rìsgənán 'he didn't eat it'
 3SG NEG eat NEG
- /r im ed^yaaw/ rìmEd^yáw 'he arrived'
 3SG CONJ arrive
- /r im asow yaav/ rìmasOw^yáyáv 'he went north'
 3SG CONJ go north

However, when these morphemes containing long vowels are concatenated, the second long vowel in the sequence is stressless.

- (82) /r is ed^yaaw aan/ rìsEd^yáwan
 3SG NEG arrive NEG
 'he didn't arrive'
- /r im ed^yaaw yaav/ rìmEd^yáw^yav
 3SG CONJ arrive north
 'he arrived in the north'

When three long vowels are concatenated, only the second is unstressed.

- (83) /r is ed^yaaw yaav aan/ rìsEd^yáw^yaván
 3SG NEG arrive north NEG
 'he didn't arrive in the north'

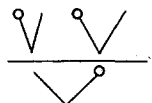
To account for this, I propose that binary left-headed ROB feet are constructed left to right prior to normal stress assignment. This converts the first example in (82) and (83) into the following.

- (84) rìsEd^yaawaan rìsEd^yaaw^yaavaan
-

Introducing a new foot type to account for consecutive syllables containing long vowels seems rather bold, given the evidence presented so far, and it would therefore be wise to consider alternatives before taking this step. The most reasonable one would be to suggest that there is a rule of shortening preceding stress assignment. Forms like

rised^yáwan would then undergo shortening (of the ultima) prior to stress assignment. This proposal entails derivations like the following.

$$(85) \quad \text{rised}^y\text{aawaan} \rightarrow \text{rised}^y\text{aawan} \rightarrow \text{rised}^y\text{aawan}$$



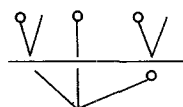
First shortening would apply to the ultima. Quantity-sensitive stress assignment then produces the correct output.

The shortening rule presumably would shorten a vowel when it immediately follows another long vowel. An approximate formulation is given in (86).

$$(86) \quad V \rightarrow [-\text{long}] / [+ \text{long}] C_0 ___$$

This rule when applied to forms like /risedyaawyáwaan/ gives the wrong results if applied in simultaneous SPE fashion. This is demonstrated in the following derivation.

$$(87) \quad \text{rised}^y\text{aawyáwaan} \rightarrow \text{rised}^y\text{aawyávan} \rightarrow * \text{rised}^y\text{áawyávan}$$



First, shortening scans the span identifying all potential foci of the rule. Then, in conformity with simultaneous application, these foci undergo the structural change producing the intermediate form above. That, in turn, undergoes the stress rules. (These are quantity-sensitive, and since this is a verb, bidirectional.) First, a left-headed foot is built on the right. Then, the rest of the domain is filled with left-headed trees built from the left. These are incorporated into a right-headed word tree. Also since long vowels do not undergo Destressing, the antepenult retains stress. To avoid the incorrect output above, we must assume that shortening either applies directionally from left to right, or that it applies cyclically. Applying the rule in either fashion produces the correct output.

Shortening appears to be a reasonable alternative to the introduction of a new foot type. However, as we will show below, shortening cannot account for the full range of long syllable cases. The problem is that the shortening analysis requires that long vowels be stressed by the same footing rules that stress short vowels. The previous ROB analysis has long vowel footing occurring prior to regular footing.

Consider now the full interaction of the verbal stress rules with a final

long vowel. Recall that in verbal stressing one regular foot is built from the right edge. If the ultima is long, then we have a test case for the two approaches sketched above. Under the shortening analysis, given in (88), we would expect that foot to dominate the long vowel. Under the ROB analysis in (89), we would expect it to occur to the left of the ROB foot which will dominate the long vowel.

(88) *Stressing verbs with a shortening rule*

- a. Apply shortening (86).
- b. Build one quantity-sensitive left-headed foot on the right edge of the word.
- c. Build quantity-sensitive left-headed feet from left to right.
- d. Build a right-headed word tree.

(89) *Stressing verbs with ROB feet*

- a. Build left-headed binary ROB feet from left to right (feet where heads *must* branch, and nonheads *may* branch).
- b. Build one left-headed binary foot on the right.
- c. Build left-headed binary feet from the left.
- d. Build a right-headed word tree.

Now consider the following verbs.

(90) áam

'to see'

r ìm áam

3SG CONJ see

'he saw it'

r ìm am áam

3SG CONJ CONT see

'he was looking at it'

t ìn ag am w áam

FUT 2 CUR CONT DU see

'you two will see it'

t ìn ag àm w abgéén

FUT 2 CUR CONT DU jealous

'you two will be jealous'

nà d y ag am ày bagéén

about-to FUT 1PLEX CUR CONT PL jealous

'we are going to be jealous'

t y Ès am àr OlgÈygEy áan
FUT 1PLEX NEG CONT PL like NEG
 'we won't like it'

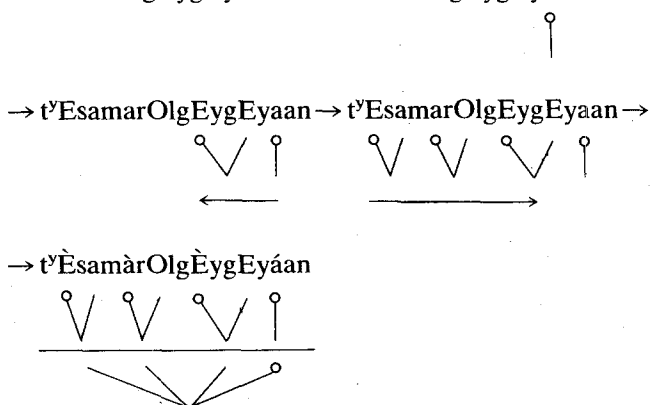
nà d y Es àm ar OlgÈygEy áan
about-to FUT 1PLEX NEG CONT PL like NEG
 'we aren't about to be liking it'

In all forms the final long vowel receives primary stress. To test the two analyses in (88) and (89), we need to determine whether the foot to the immediate left of this foot is built from the right or from the left. If this foot is built from the left, then in even-syllabled words it should be degenerate. That is because after subtracting the long vowel, an odd number of syllables remain. If the foot preceding the long vowel is built from the right, then, since its construction precedes left-to-right foot construction, it will be binary in even-syllabled spans longer than two. In such cases, it will be preceded by a degenerate foot. This is because subtracting the long vowel and the preceding two syllables from an even-syllabled word leaves an odd-syllabled span.

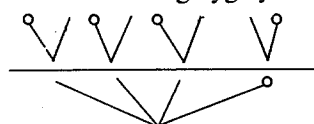
The relevant forms in (90) are the four-, six-, and eight-syllabled cases. Since degenerate feet are in the normal case removed by Destressing, the relevant evidence will be a gap. That is, rather than finding a degenerate foot in the locations discussed above, we should look for a gap (two stressless syllables) in the locations discussed.

In the six- and eight-syllabled cases, the gap occurs in the position predicted by the ROB analysis: before the penultimate stress. In (91) and (92) below are derivations of the seven- and eight-syllabled cases in terms of that analysis.

(91) tʰEsamarOlgEygEyaan → tʰEsamarOlgEygEyaan →



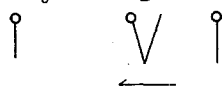
→ *nàdʷEsàmarÒlgEygEyáan



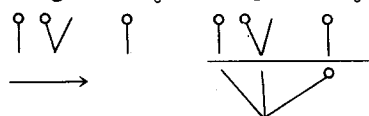
Shortening would apply first. Since there is only one long vowel in this form, it is inapplicable. Next, a quantity-sensitive foot is built from the right. Since there are no ROB feet in this analysis, that first foot ends up degenerate, dominating the long vowel. Then, more feet are built from left to right. Notice that this now results in a degenerate foot on the penult since the remaining string has an odd number of syllables. That foot would eventually undergo Destressing giving the wrong result exemplified. Hence, the ROB analysis gives the right results while the OB analysis with Shortening does not.

Consider now the four-syllabled case. Assuming the ROB analysis, this should undergo the derivation below:

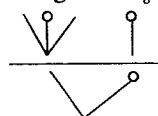
(94) t̃inagamwaam̩ → N/A → t̃inagamwaam̩ → t̃inagamwaam̩ →



→ t̃inagamwaam̩ → t̃inagamwaam̩ → Destressing →



→ *t̃inàgamwám̩



As the derivation above shows, our analysis so far predicts antepenult secondary stress rather than preantepenult secondary stress in this case.

To get the right results in a case like this, we must remove the penultimate foot rather than the antepenultimate foot. The rule below will accomplish this.

(95) [v̩ o̩ o̩] ... ⇒ [v̩ o̩] ...

This rule prunes a binary foot not the head of the word tree after an initial degenerate foot in a verb. This rule would have to be ordered before the general destressing rule (78) above.

This may seem ad hoc, but on closer inspection, it is quite natural.

First, the stipulation that the first foot be degenerate and the second binary is just an indirect way of appealing to the notion "stress clash" which Hammond (1984) argues is central to rhythm and destressing rules. The stipulation that the second foot not bear main stress is also a familiar constraint on destressing (Hayes, 1981; Prince, 1983; Hammond, 1984). Lastly, the restriction that the rule apply only to verbs can perhaps be related to the following observation: in a stress clash involving no main stress, the foot occurring at its own origin of construction is not the foot destressed. By 'origin of construction' is intended the left edge in left to right footing and the right edge in right to left footing. The verbal restriction of the rule is a way of indicating that the degenerate foot is at its origin of construction, in this case, the left edge of the word. This entails that it cannot be destressed if the observation above is correct – because it is only in verbs that left to right foot-construction occurs. A preliminary investigation of all destressing rules in Hammond (1984), Hayes (1981), and Prince (1983) reveals no counterexamples to that (admittedly tentative) observation.

Another possible analysis of the facts presented makes use of OB. This is essentially the proposal made by Lynch (1974, 1978) and by Halle and Vergnaud (1985) translated into the theory presented here. Halle and Vergnaud propose the analog of a cyclic rule of shortening followed by their analog of OB. As in the ROB analysis presented, OB footing precedes normal footing, and the facts of (90) follow directly. Empirically, this proposal can be distinguished from the current one if there are any prefixes containing long vowels. If there were, then cyclic shortening would result in all but the first such prefix in sequence being shortened. A second possible empirical difference concerns the applicability of laxing to long vowels in sequence. The cyclic shortening analysis (potentially) predicts that shortened vowels would undergo laxing. The ROB analysis predicts not. Lastly, the cyclic shortening analysis is contingent on the cyclicity of the stratum on which long vowel suffixes are added. If that stratum were noncyclic, then the shortening analysis would not work.

Unfortunately, the data available do not discriminate between the two analyses. Thus, in the case of Lenakel, our choice between them must so far be made on theoretical grounds. Since ROB is conceptually superior to OB, there is at least one argument in favor of the ROB analysis. Moreover the ROB analysis requires no shortening rule. In the absence of evidence supporting such a rule, it must be viewed as an unnecessary complication, required only by the OB analysis. We now turn to a discussion of lexically stressed syllables in Lenakel. By showing that these exist, have different properties from long vowels, and are more properly

represented as lexically stressed, we make the case tighter that long vowels are stressed by a different mechanism: ROB feet.

Lynch cites the transitive suffix **-in** as being preceded by the boundary “=”. His motivation for this includes the following facts. First, it surfaces as stressed even when it is the ultima.

- (96) $\text{h}\dot{\text{ı}}\text{nad}^{\text{y}}=\dot{\text{ı}}\text{n}$ ‘to know’
 $\text{ə}\eta\text{n}=\dot{\text{ı}}\text{n}$ ‘to be afraid of it’
 $\text{à}\text{řu}=\dot{\text{ı}}\text{n}$ ‘to throw’
 $\text{á}\text{ğle}=\dot{\text{ı}}\text{n}$ ‘to steal it’

In the examples above, one would expect the normal penultimate stress pattern. A second fact about **-in** is that it may surface with a lax vowel if the phonological conditions of laxing are satisfied, as they are in all the examples above. Recall that long vowels always surface as tense. Third, Lynch shows that a sequence of two long vowels is stressed differently from a sequence of long vowel followed by **-in**. Contrast the following examples. In the first one, the last two (long) vowels are footed together by ROB footing. This results in the ultima being stressless. In the second example, the suffix **-in** bears stress, showing that it is not footed by ROB footing.

- (97) /r əm edyaaw yaav/ [rəmEd^yáw^yav]
 3SG CONJ arrive north
 ‘he arrived in the north’

 /r alak yaav =in/ [rələk^hyávIn]
 3SG drop north TRANS
 ‘he dropped it down’

These three facts indeed suggest that **-in** should be treated differently from the long vowel cases. However, Lynch’s boundary solution merely stipulates diacritically that **-in** has certain exceptional properties. Each rule in Lynch’s analysis must then include the boundary symbol. What this means is that the stress rule and the destressing rule must both include mention of the boundary so as to account for the properties discussed above. Under such an analysis, there is no intrinsic connection between the two rules. That is, there is no reason why these two rules alone refer to the boundary. In light of this I propose an alternative treatment for **-in**: that it is lexically stressed. That is, its lexical representation is as in (98).

first, bleeding (78). Both rules apply so as to eliminate clashes, a central claim of Hammond (1984) and Prince (1985). Moreover, (95) seems to be a consequence of a principle whereby a foot at its own origin of construction is relatively resistant to destressing. It might be objected that it is dubious to propose theoretical innovations on the basis of exceptional morphemes.¹⁶ However, it is not just the behavior of exceptions that has prompted the proposal, but the interaction of exceptions with the productive stress assignment algorithm.¹⁷

5. KLAMATH

The stress system of Klamath provides interesting support for the revision of OB suggested here.¹⁸ The facts can be summarized as follows.

- (100)a. The rightmost long vowel receives primary stress;
- b. otherwise, stress a closed penult;
- c. otherwise, stress the antepenult if there is one;
- d. otherwise, stress the penult.

Data exemplifying each case are provided below.

- (101) *long vowel*
 naqá:qbli 'puts a flat object on one's lap'
 sʔo:dí:la 'puts a tray of food under'
 čatá:wipga 'is sitting in the sun'
- (102) *closed penult*
 gatbámbli 'returns home'
 qmaqkánga 'looks around for'
 taktákli 'red'
- (103) *antepenult*
 cáwiga 'is crazy'
 ʔápʔota 'promises'

¹⁶ Thanks to an anonymous reviewer for drawing this point to my attention.

¹⁷ In an earlier discussion of ROB and OB, I argued that main stress in Cairene Arabic can be accounted for using ROB feet. If the secondary stress facts are as described in Welden (1980) and Harms (1981), then that account goes through. Unfortunately, the secondary stress facts are dubious. If one ignores the distribution of secondaries, then the facts are equally well described using OB feet (Hammond, 1985). Thanks to an anonymous reviewer for sharing his fieldwork with me on this issue.

¹⁸ The data are from Barker (1963, 1964). Thanks to J. Levin for drawing my attention to the relevance of Klamath for revised OB. See Levin (in prep.) for an alternative analysis.

- (104) *penult*
 bóčo ‘wild celery’
 káni ‘who?’
 glégatk ‘dead?’

Consider first the data in (101). To account for these one needs either quantity-sensitive feet or ROB feet. If the feet were quantity-sensitive, then they would have to be left-headed, because it is the rightmost long vowel that bears the main stress. Recall that a left-headed quantity-sensitive foot seeks out the rightmost heavy, while a right-headed quantity-sensitive foot picks out the leftmost heavy. If the feet were left-headed quantity-sensitive feet, then we would expect initial stress just in case the word contained no long vowel, since a quantity-sensitive foot may assign stress to a light syllable, and, if there are no long vowels in the span, maximality of foot construction will force a left-headed foot to place initial stress. The data in (102) show that this is incorrect. Therefore the feet are not quantity-sensitive, but ROB.

If the feet are ROB, then to stress the rightmost long vowel, a right-headed ROB foot must be built. (Notice that this is the mirror image of the rule for Khalkha in (45a).) In (105), we give this preliminary analysis.

- (105) *Preliminary Analysis*
 Build a right-headed ROB foot.

This accounts directly for the words in (101).

- (106) sʔo:dí:la čatá:wipga
 V V

However, (105) leaves all other cases – (102), (103), and (104) – unstressed.

Before revising (105) to account for these, let us consider secondary stress assignment. Barker (1964) cites the following examples.

- (107) gabá:tàmbli ‘goes back to shore’
 gawí:napgàbli ‘is going among again’

According to Barker, secondary stress falls on the penult when “either the penult or the final syllable are closed and high pitch and stress occur upon a long vowel preceding the penult” (p. 36). Translated into more theory-dependent terminology, this means that if the ROB foot is built, then secondary stress may occur to the right on a penult if it is heavy *or*

the ultima is heavy. What is odd about this is that stress should be attracted to a syllable if a neighboring syllable is heavy.

Levin (personal communication) observes that examples of light penults that should receive secondary stress because of a heavy ultima are extremely rare. A search through most of the *Klamath Dictionary* (Barker, 1963) revealed no more than thirty examples. These can be divided into four types of cases. The largest class is composed of plural kinship terms. These all occur with the plural marker **-is** followed by the kinship marker **-ap**. Many also occur with the kinship prefix **p-**.

- | | | |
|------|--------------|---------------------------------|
| 108) | pqe:sisap | 'sons-in-law' |
| | p'a:ktisap | 'nephews, nieces' |
| | mamgo:q'isap | 'nephews, nieces' |
| | tge:wn'isap | 'elder brothers, sons, cousins' |
| | pt'e:wisap | 'grandmothers, grandchildren' |
| | pse:ysisap | 'uncles, nephews, nieces' |
| | psa:sisap | 'stepmothers, stepchildren' |
| | psa:q'isap | 'aunts, nephews, nieces' |
| | p?a:nisap | 'woman's elder brothers' |

A second (potentially) large class involves the possessive allomorph **-am**. This seems to occur only with plural animates.

- | | | |
|-------|----------|-------------------------------------|
| (109) | ge:yasam | 'their (proximate)' |
| | ne:yasam | 'their (absent)' |
| | p?anisam | 'woman's elder brothers (genitive)' |

A third category includes species of trees marked with the homophonous suffix **-am**

- | | | |
|-------|-----------|------------------|
| (110) | wa:ko?am | 'lodgepole pine' |
| | qde:Lo?am | 'sugar pine' |

Lastly, there are a few odd examples that don't fall into a neat category.

- | | | |
|-------|------------------|-----------------------------|
| (111) | na:nok'ant | 'in all' |
| | mo:watwa:syalank | 'in the Pit River language' |
| | č'a:si?am'č | 'big old skunk' |

Barker does not mark stress in his dictionary, but if the generalization above is correct, we would expect secondary stress on the penult in all the examples given. For example:

- | | |
|-------|------------|
| (112) | pqé:sisap |
| | gé:yàsam |
| | wá:kò?am |
| | ná:nòk'ant |

Assuming that these are indeed the facts, how are we to account for them? There are two possibilities. One is to complicate the stress assignment algorithm so that a light penult can attract stress if it is followed by a heavy syllable. This would allow us to account for the twenty to thirty examples that might occur. However, it is not clear what the stress facts for these cases are, since no examples are stress-marked and there only appear to be a few examples. Moreover it would complicate the theory to allow these facts to be captured as part of normal stress assignment.

A second possibility would be to mark relevant forms lexically. Such lexical markings could be of two sorts. The examples in (111) could be marked as exceptions, with their stress patterns given in the lexicon. The forms in (108), (109), and (110) appear to fit within a lexical generalization to the effect that certain suffixes make the preceding syllable heavy, namely **-ap**, **-am₁**, **-am₂**. If these suffixes are marked for this property somehow, then we would have to countenance far fewer exceptions than initially appeared. By marking such cases as exceptions (of either sort), the following advantages would result: (1) we would not have to complicate the stress assignment algorithm to account for cases that we are unsure of; (2) we would not have to justify the complication in the face of only thirty examples; and (3) the theory itself would not have to be complicated.

We therefore treat these examples as exceptions of two types. Examples like (111) are lexically stressed. The other cases result from application of a rule making penults heavy before certain suffixes. Rule (113) seems adequate for this purpose:

$$(113) \quad \begin{array}{c} \sigma \\ | \\ V \end{array} \quad \begin{array}{c} \sigma \\ / \quad \backslash \\ C \quad] \left\{ \begin{array}{cc} a & m \\ a & p \end{array} \right\}] \end{array} \Rightarrow \begin{array}{c} \sigma \\ / \quad \backslash \\ V \quad C \end{array} \quad \begin{array}{c} \sigma \\ / \quad \backslash \\] \left\{ \begin{array}{cc} a & m \\ a & p \end{array} \right\}] \end{array}$$

This rule makes the onset of a suffix **-am** or **-ap** into the coda of a preceding syllable. It has the effect of making the penult heavy. If this rule precedes stress assignment, then it will enable words with superficially light penults to be treated as if they were heavy. In this way, these heavy ultima cases can be assimilated to the heavy penult cases.

This rule makes the prediction that other rules sensitive to syllable structure will treat these penults as heavy. If these rules do not, in fact, do so then the rule above can be replaced with one marking the penult [+H] or 'accented'. Depending on which version of metrical theory is adopted, these diacritics enable the relevant syllables to be treated as heavy syllables.

Given this, we assume that the generalization covering secondary stress is as follows.

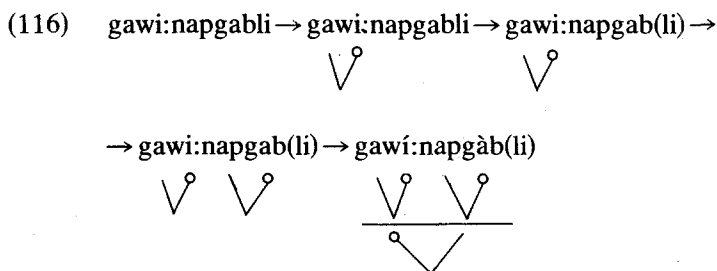
- (114) Secondary stress falls on a closed penult if main stress occurs on a long vowel to the left.

This simplifying assumption does not alter the relevance of Klamath for revised OB.

To account for secondary stress, we need to build a second ROB foot in addition to the one built by (105). This foot, however, differs from the first in that (1) it is sensitive to the distinction between light syllables and closed syllables; and (2) this foot only recognizes this distinction when it involves the penult. To make the distinction appropriately, the final syllable could be made extrametrical, and the second ROB foot built on the right edge of the word. A partial analysis incorporating these ideas is given below.

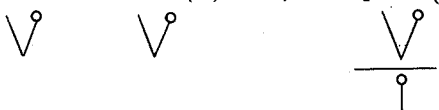
- (115)a. Build an unbounded right-headed ROB foot, where the head must dominate a long vowel, and the nonheads may dominate any kind of syllable.
 b. Make a final unstressed syllable extrametrical.
 c. On the right edge of the word, build an unbounded right-headed ROB foot, where the head must dominate a heavy syllable, and the nonheads may dominate any kind of syllable.
 d. Build a left-headed word tree.

This system provides analyses like the following for words with secondary stress.



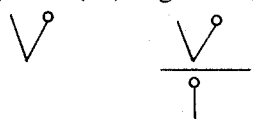
First, an ROB foot sensitive to the distinction long/short is built. Then the final vowel is made extrametrical, and another ROB foot, this time sensitive to the light/heavy distinction is built on the right edge. Lastly, the output is incorporated into a left-headed word tree. This analysis also

accounts for long vowel cases without secondary stress, e.g.:

$$(117) \quad s?o:di:la \rightarrow s?o:di:la \rightarrow s?o:di:(la) \rightarrow N/A \rightarrow s?o:dí:(la)$$


First, the long vowel ROB foot is built, and then extrametricality applies. Here, the second ROB foot cannot be built, since the span is completely covered. The word tree is added (vacuously).

This analysis has the nice property of accounting for the cases in (102), e.g.:

$$(118) \quad gatbambli \rightarrow N/A \rightarrow gatbam(bli) \rightarrow gatbam(bli) \rightarrow gatbám(bli)$$


Long vowel ROB is inapplicable, and extrametricality applies feeding closed syllable ROB footing. The word tree is added (vacuously).

The analysis also accounts for the forms in (104). For example:

$$(119) \quad bočo \rightarrow N/A \rightarrow bo(čo) \rightarrow N/A \rightarrow bó(čo)$$


Neither case of ROB applies, and extrametricality feeds word tree construction.

In fact, though it is not in accord with Barker's description, the analysis given so far will also account for the examples in (103).

$$(120) \quad ?ap?ota \rightarrow N/A \rightarrow ?ap?o(ta) \rightarrow N/A \rightarrow ?áp?o(ta)$$


Neither ROB footing applies, and as with **bóčo**, the word tree places the correct stress.

Although Barker does not provide the stress, his description would lead us to believe that a word of at least four syllables with no long vowel and a light penult should get antepenult stress.

$$(112) \quad (C_0V)_1C_0\acute{V}C_0VCVC_0$$

In such a case, our analysis predicts initial stress. In fact, a search through Barker's *Klamath Dictionary* reveals a few hundred exam-

ples that would test this. In (122) several are given, marked with the stress Barker's description suggests.

- (122) papákpak'a 'scream'
 gemblinánnwiwapk 'will go right back'
 yebántq'aga 'digs something up'

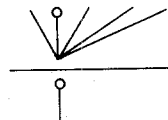
Since there are hundreds of such examples, marking them as exceptions to the analysis in (115) is much less palatable than was the case for the examples discussed earlier.

If they are to be accounted for using the core stress assignment algorithm, then what we seem to need is a 'binary word tree'. A binary word tree would take syllables as terminals and seek out the antepenult, since the ultima is extrametrical. Such an account would force us to enrich the theory to allow binary word trees – an undesirable move. However, an alternative is available. Stowell (1979) has argued that there is an intermediate footlike structure between feet and word trees: superfeet. Such structures operate on the output of foot construction, binding feet together pairwise to produce four degrees of stress when the word tree is added. Stowell's proposal is motivated on the basis of a 'binary' pattern of alternation in Passamaquoddy where, ideally, every *fourth* vowel gets secondary stress.

Let us assume then that in Klamath, word tree construction (now unnecessary) is preceded by cohstruction of a left-headed binary superfoot. Such a structure would produce the right outputs when ROB feet are built, since the leftmost stress is primary, and there are never more than two feet in a word. However, when no feet can be built, the binary superfoot takes syllables as terminals and places stress on the antepenult instead of the initial syllable. This move is in accord with Barker's description, and involves no extensions to metrical theory, unlike the binary word tree approach.

Summarizing, ROB feet in Klamath are *right-headed*. This permits secondary stress to occur to the right of the long vowel foot. Using unrevised OB feet, there would be no way to place secondary stress since the OB feet would have to be *left-headed*, leaving no room to the right for the secondary stress foot, e.g.:

- (123) gawi:napgabli → gawi:napgabli → N/A → *gawí:napgabli



First, the left-headed OB foot is built. There is now no way to build a secondary stress foot on the right edge of the word since the right edge is dominated by the left-headed foot. Thus the analysis I have given argues for ROB feet, since they allow a natural treatment of secondary stress. If, as shown above, OB feet were constructed, there would be no way to place secondary stress to the right of the primary.

The analysis makes general predictions about the possible distribution of secondary stress. In a system like Klamath where the ROB feet are right-headed, secondaries can occur to the right of the primary. In an ROB system where feet were left-headed, secondaries would only be able to occur to the left of the primary. These predictions follow from the constituency entailed by an analysis in terms of metrical trees. An analysis in terms of a grid would not make any predictions with regard to secondaries. In this way, the ROB analysis of languages like Klamath fits into a more restrictive typology than a grid-based account of the same facts could.

6. CONCLUSION

In the preceding sections, I have argued that metrical theory should be revised to incorporate revised obligatory-branching feet. This argument was based on the descriptive necessity of bounded ROB feet for Lenakel and unbounded ROB feet for Klamath. Neither of these languages are describable as simply using unrevised OB feet.

One nice consequence of ROB footing is that it allows us to constrain the types of restrictions that are put on feet. Quantity-insensitive foot construction puts no constraints on what may qualify as terminals. Quantity-sensitive foot construction limits nonheads of feet. ROB foot construction limits heads of feet. Feet can therefore have unconstrained terminals, or one type of terminal, either heads or nonheads, can be constrained. Old-style OB feet constrained both head and nonhead, and one was left with a curious gap. There were no foot types where only the heads were restricted. Now, by adopting ROB feet, the gap is much more reasonable. Feet where both head and nonhead are constrained do not occur.

What have we learned about stress? What insight is expressed by ROB feet? In one sense, this is a meaningless question. ROB feet are a descriptive device that accounts for a range of otherwise difficult to account for facts. In another sense, though, theoretical devices may express some insight, that is, satisfy linguists conceptually in addition to accounting for the data in as simple a fashion as possible. If so, then it is

reasonable to ask what intuition is captured by ROB feet.¹⁹ The intuition is that although heavy syllables tend to attract stress, they do not always do so. In particular, ROB feet enable us to capture the fact that in certain positions (nonhead positions) under certain types of iteration or scanning (ROB types), heavy syllables are treated just like light syllables. With ROB feet, heavy syllables just like light syllables can be nonheads and hence may be unstressed. In this light, ROB feet can be seen as intermediate between quantity-insensitive and quantity-sensitive feet. Quantity-insensitive feet scan ignoring syllable weight. Quantity-sensitive feet scan allowing heavy syllables to interrupt the iteration at any time. ROB feet allow heavies to interrupt the iteration some of the time. Thus, ROB should not be seen as a bizarre foot type tacked onto the system, but as a natural, expected foot type intermediate between quantity-sensitivity and quantity-insensitivity and expressing the same basic intuitions about stress systems.

REFERENCES

- Al-Mozainy, H. Q., R. Bley-Vroman, and J. McCarthy: 1985, 'Stress Shift and Metrical Structure', *Linguistic Inquiry* 16, 135-144.
- Anderson, S. R.: 1984, 'A Metrical Interpretation of Some Traditional Claims about Quantity and Stress', in Aronoff, M. and R. T. Oehrle (eds.), *Language Sound Structure*, MIT Press, Cambridge.
- Barker, M. A. R.: 1963, *Klamath Dictionary*, University of California Press Publication in Linguistics 31, Berkeley.
- : 1964, *Klamath Grammar*, University of California Press Publication in Linguistics 32, Berkeley.
- Benger, J.: 1984, *The Metrical Phonology of Cayuga*, M.A. Linguistic Forum, University of Toronto, Toronto.
- Bolozky, S.: 1982, 'Remarks on Rhythmic Stress in Modern Hebrew', *Journal of Linguistics* 18, 275-289.
- Chomsky, N. and M. Halle: 1968, *The Sound Pattern of English*, Harper & Row, New York.
- Crowley, T.: 1978, *The Middle Clarence Dialects of Bandjalang*, Australian Institute of Aboriginal Studies, Canberra.
- Halle, M. and J.-R. Vergnaud: 1978, 'Metrical Structures in Phonology', manuscript, MIT.
- : 1985, 'Stress and the Cycle', manuscript.
- Hammond, M.: 1984, *Constraining Metrical Theory: A Modular Theory of Rhythm and Destressing*, Ph.D. dissertation, UCLA, revised version distributed by Indiana University Linguistics Club.
- : 1985, 'Obligatory Branching Revisited', to appear in the proceedings of *NELS* 15.
- Hansen, K. C. and L. E. Hansen: 1969, 'Pintupi Phonology', *Oceanic Linguistics* 8, 153-170.
- Harms, R. T.: 1981, 'A Backwards Metrical Approach to Cairo Arabic Stress', *Linguistic Analysis* 7, 429-450.

¹⁹ Thanks to an anonymous reviewer for reminding me of this.

- Hayes, B.: 1981, *A Metrical Theory of Stress Rules*, Ph.D. dissertation, MIT, revised version distributed by Indiana University Linguistics Club.
- : 1984, 'The Phonology of Rhythm in English', *Linguistic Inquiry* 15, 33–74.
- : 1985, 'Iambic and Trochaic Rhythm in Stress Rules', *Berkeley Linguistic Society* 11, 429–446.
- Kaisse, E.: 1985, 'Some Theoretical Consequences of Stress Rules in Turkish', to appear in the proceedings of the Chicago Linguistics Society.
- Levin, J.: 1982, 'Metrical Structure in Cahuilla', manuscript, MIT.
- : in preparation, 'The X-tier and Klamath Phonology', manuscript, MIT.
- Lieberman, M. and A. Prince: 1977, 'On Stress and Linguistic Rhythm', *Linguistic Inquiry* 8, 249–336.
- Lynch, J.: 1974, *Lenakel Phonology*, Ph.D. dissertation, University of Hawaii, in *University of Hawaii Working Papers in Linguistics* 7.
- : 1978, *A Grammar of Lenakel*, Pacific Linguistics, Series B, No. 55, Australian National University.
- Prince, A.: 1983, 'Relating to the Grid', *Linguistic Inquiry* 14, 19–100.
- : 1985, 'Improving Tree Theory', *Berkeley Linguistic Society* 11, 471–490.
- Rappaport, M.: 1984, *Issues in the Phonology of Tiberian Hebrew*, Ph.D. dissertation, MIT.
- Sebeok, T. A. and F. J. Ingemann: 1961, *An Eastern Cheremis Manual*, Indiana University Publications, Bloomington.
- Selkirk, E. O.: 1984, *Phonology and Syntax*, MIT Press, Cambridge.
- Sezer, E.: 1983, 'On Non-final Stress in Turkish', *Journal of Turkish Studies* 5, 61–69.
- Stowell, T.: 1979, 'Stress Systems of the World, Unite!', in *Papers on Syllable Structure, Metrical Structure, and Harmony Processes*, (K. Safir, ed.) *MIT Working Papers in Linguistics* 1.
- Street, J. C.: 1963, *Khalkha Structure*, Indiana University Publications, Bloomington.
- Underhill, R.: 1976, *Turkish Grammar*, MIT Press, Cambridge.
- Welden, A.: 1980, 'Stress in Cairo Arabic', *Studies in the Linguistic Sciences* 10, 99–120.

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Dept. of Linguistics
Univ. of Wisconsin – Milwaukee
Milwaukee, WI 53201
U.S.A.